

Demonstrating Hapticus: Exploring the Effects of Haptic Feedback and its Customization on Motor Skill Learning: Tactile, Haptic, and Somatosensory Approaches

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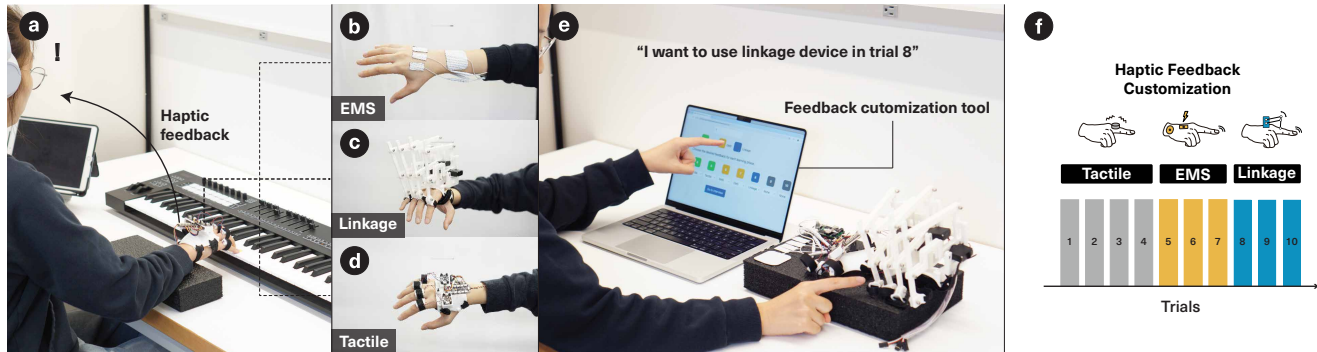


Figure 1: (a) We explored the motor learning performance and experiences with different haptic feedback devices during piano learning. We prepared (b) a hand exoskeleton with mechanical linkages (linkage), (c) a vibrotactile glove (tactile), and (d) a system using Electrical Muscle Stimulation (EMS). (e) We then examined whether the customization of the haptic feedback during motor learning can further improve the learning experience, (f) demonstrating better learning performance and experience in the customization condition.

Abstract

Numerous haptic devices have been proposed to support motor learning. Understanding the impact of each type of feedback on users' learning performance and experience, as well as the effects of customizing the haptic feedback each user receives, is vital to achieving both efficient and highly motivating learning. To explore this, we developed three devices to assist piano learning: a hand exoskeleton with mechanical linkages (linkage), a vibrotactile glove (tactile), and Electrical Muscle Stimulation (EMS). Furthermore, we demonstrate a haptic customization, where users can customize the feedback order based on their preferences.

CCS Concepts

• Human-centered computing → Haptic devices.

Keywords

motor skill learning, electrical muscle stimulation, exoskeleton, vibrotactile

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1 Introduction

Motor skill learning is essential for performing daily activities [1, 6]. A wide range of motor skills are integral to everyday life, from basic tasks such as walking to complex tasks like surgical procedures and playing musical instruments. However, developing dexterous motor skills is challenging to achieve independently [2, 6] as motor skill learning requires mastering multidimensional aspects, including the precise application of force [17], dexterity [20], timing [3] and directional control [26]. For instance, not only does a piano learner need to understand the musical notes and sequences, but also physically coordinate their fingers to apply the correct amount of force at the right times, on the right keys.

Although other forms of fine-motor skill improvement devices have been used, such as visually providing related body movements by using flat screens [12], projectors [19], and head-mounted displays [10] from one to another user. While visual feedback is excellent at demonstrating where and how body parts should move, it cannot directly convey essential elements of motor skills, such as muscle force or coordination.

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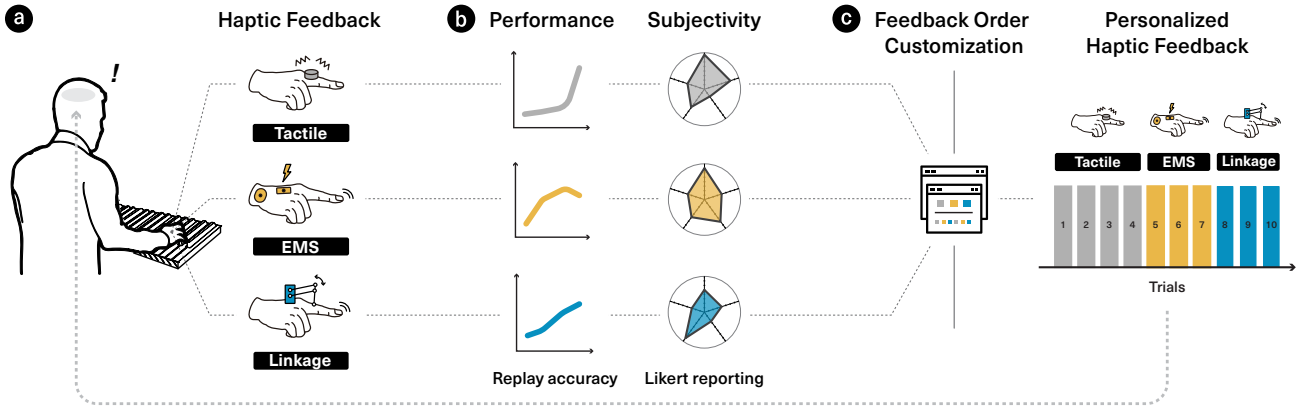


Figure 2: (a) We compared how three different haptic modalities (*i.e.*, a vibrotactile glove (tactile), EMS, and a hand exoskeleton with mechanical linkages (linkage)) affect motor learning (b) In visitor experience 1, we examined how each type of feedback impacts learning performance and subjective experiences during piano learning. (c) In visitor experience 2, we explored the effects of haptic customization, where users can customize the feedback order based on their preferences.

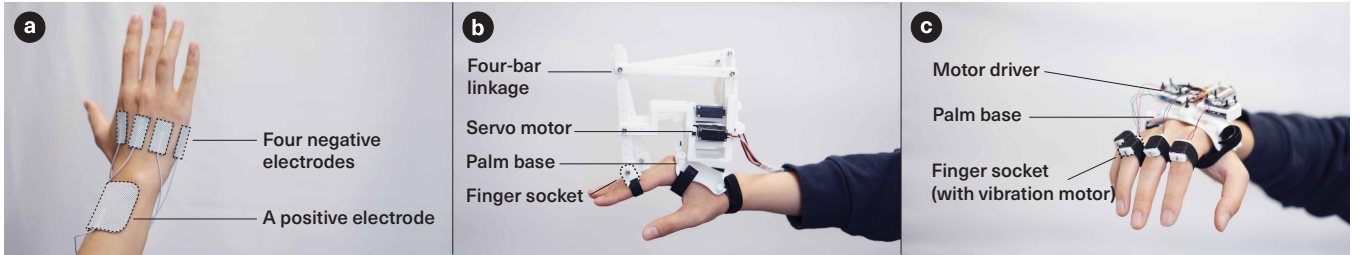


Figure 3: (a) The EMS device includes four negative electrodes attached to the back of the hand and a positive electrode on the wrist. (b) The linkage device consists of a palm base and four linkages. A linkage consists of a four-bar linkage, a servo motor, and finger sockets. (c) The tactile device features a palm base, motor drivers, and finger sockets equipped with four vibrotactile motors.

To address this, haptic feedback systems have been proposed. Examples include the use of vibrotactile motors [9], mechanical linkages exoskeletons [23], and electrical muscle stimulation (EMS) [5] that can physically actuate muscles through electrodes attached to the skin. Haptics that directly affect the physical actions of a user have proven to be effective in scenarios that require the learning of fine motor movement such as playing musical instruments [23], sports skills [11], or rehabilitation [7, 18]. These systems demonstrated a number of positive effects on motor learning efficacy, such as improved skill performance [15, 23] and limb movements [16, 25], and a more engaged experience [13, 24].

To explore the characteristics of each haptic modality and their respective effects on motor learning performance and experience, we prepared three different haptic devices that can be used for piano learning – a vibrotactile glove (tactile), EMS, and a hand exoskeleton (linkage) (Figure 2a and b). Furthermore, we introduce a feedback order customization platform (Figure 2c) along with a combined haptic device where users can customize the feedback order based on their preferences.

2 Implementation: Wearable Piano-skill Feedback Systems

To demonstrate the effects of different haptic on motor skill learning, we developed three wearable devices as shown in Figure 3: a vibrotactile glove (tactile), EMS, and an exoskeleton with mechanical linkage (linkage) which have been commonly used in motor learning research. These devices provide feedback to four fingers, excluding the thumb, due to the technical limitations of EMS actuation [22]. The devices used in this study are as follows:

2.1 EMS

The EMS device consists of five electrodes and patches. The patches, each measuring 1 x 3 cm, are placed on the user's back of the right hand with negative electrodes and wrist with a positive electrode as shown in Figure 3 (a). The device provides an adjustable pulse width from 0 to 300 μ s and operates at a stimulation frequency of 100Hz. It consists of a microcontroller (Arduino, Nano 33 BLE), gate-switching devices, and DC-DC converters generating an output amplitude of 35V from AAA batteries. These specifications are consistent with bioSync [14], and have previously been utilized in

related motor learning research [4, 5, 21]. We mainly referenced a previous study on EMS finger actuation [22], including aspects such as patch placement and size. The weight of this device is 20g. The average actuation latency from stimulus to playing note is 32 ms (SD = 8.3 ms).

2.2 Exoskeleton with Mechanical Linkages

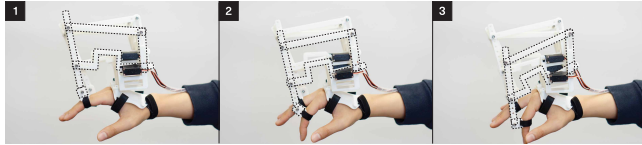


Figure 4: Linkage actuation mechanism: Our linkage device was developed to actuate four individual fingers (index, middle, ring, and pinky), enabling precise downward motion for note-playing.

We designed and implemented a hand exoskeleton, as shown in Figure 3 (b) and Figure 4. The device features four four-bar linkages, servo motors (FEETECH, FT1117M), and finger sockets mounted on a palm base. The form factor of this device is 3D-printed and the weight of this device is 287g. Servo motors are controlled by a microcontroller (Arduino, Nano 33 BLE), which is powered by a 6V power supply unit. The actuated angle for the exoskeleton is set at 40 degrees. For intensity control, the speed of the motors is adjusted. The average actuation latency from stimulus to playing note is 108 ms (SD = 19.2 ms).

2.3 Vibrotactile Glove

We prepared a vibrotactile glove that is a 3D-printed hand apparatus equipped with four linear resonant actuator (LRA) motors (Vybronic, VG0832022D), which have dimensions of 8mm in diameter and 3.2mm in thickness. Each motor operates at an approximate vibration frequency of 235Hz. As shown in Figure 3 (c), these motors are securely placed in 3D-printed sockets and attached using velcro on top of each user's finger for easy adjustment and stability. The motors are controlled by a microcontroller (Arduino, Nano 33 BLE) and motor drivers (DRV2605L), powered by a pair of AAA batteries (3V). The weight of this device is 63g.

2.4 Combined Device

For usage on the second scenario during the visitor experience, and to provide seamless transitions of multiple haptic feedback, we developed an integrated device that combined EMS, linkage, and tactile actuators as shown in Figure 5 and its weight is 309g. A modification to this device was redesigning the finger fastener system. The original linkage design was revised to enable better adjustability, allowing for easier tightening and loosening of the velcro around the finger. This adjustment enables more flexible finger movement, especially when using the EMS.

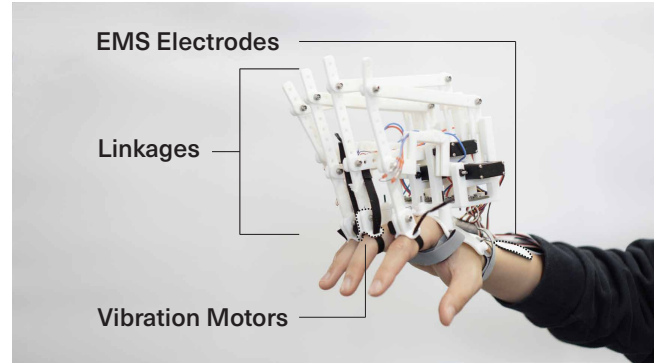


Figure 5: Haptic device used in visitor experience 2; we developed an integrated device where the participants wore EMS patches, with a linkage device positioned on top. Vibrotactile motors, located inside the finger sockets, provided tactile feedback.

3 Visitor Experience

3.1 Scenario 1: Comparing Three Haptic Feedback

The first demo scenario involves experiencing the differences between each type of feedback (i.e., EMS, linkage, and tactile) while learning a 6-note melody. Visitors can try each type of feedback, enabling them to compare their learning experience and performance across the feedback types. After completing each trial, their performance will be displayed on a monitor.

3.2 Scenario 2: Exploring Haptic Customization

Figure 6: Feedback order customization form: At the end of visitor experience 1, participants specified the preferred type of feedback to receive per trial. Recorded data is used as haptic order customization in visitor experience 2.

The second scenario focuses on haptic customization. Based on our findings [8], we discovered that customization enhances user agency and learning performance. Visitors can customize the feedback order based on their preferences using a web-based interface (Figure 6). This provides a personalized experience tailored to their choices, allowing them to evaluate how their customization affects their learning experience and performance.

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